

George Webber

📍 London, UK ✉ george.w11@hotmail.co.uk ☎ +44 7741 457 121 in [george-webber](#) 🌐 [george-webber.com](#)

PhD candidate at King's College London, specializing in inverse problems and generative machine learning for PET/MR imaging. Holds a First-Class Integrated Master's degree in Mathematics & Computer Science from the University of Oxford. Research aims to advance medical image quality through innovative integration of machine learning methods.

Education

King's College London

Sept 2023 – Present

PhD in Biomedical Engineering & Imaging Sciences

- Project: *Fully Bayesian 3D PET-MR Neuroimaging Reconstruction*.
- Investigating pre-trained diffusion models as generative priors for inverse problems in medical imaging.
- Supervisors: Prof. Andrew J. Reader and Prof. Andrew P. King.

St Anne's College, University of Oxford

Oct 2019 – Jun 2023

MMathCompSci – Mathematics & Computer Science

- First Class overall (77 %) and Distinction in final year (81 %).
- Master's thesis: *Improving the Security of Smartwatch Payment with Deep Learning*.

John Mason School, Abingdon

Sept 2012 – 2019

A-Levels and EPQ

- A*A*A*A*A* in Mathematics, Further Mathematics, Computer Science, Physics, and an EPQ.

Experience

Graduate Teaching Assistant

London, UK

King's College London

Nov 2023 – Present

- Deliver tutorials for undergraduate mathematics and labs for an interdisciplinary medical imaging module.
- Mark coursework and provide detailed feedback to cohorts of 60+ students.

Alpha Tech Intern – Quant Analysis

London, UK

Man Group

Jun – Sep 2022

- Designed experiments on simulated portfolios, enhancing Value-at-Risk calculations by incorporating earnings-announcement effects.
- Co-authored an internal research paper and delivered a 30-minute presentation to senior risk managers.
- Shipped production-ready Python code to calculate daily risk metrics.

Senior Consultant

Oxford, UK

Oxford Strategy Group Digital

Sep – Dec 2021

- Open-sourced and extended Wise's variational-autoencoder model for customer-behaviour prediction.
- Authored tutorials and Jupyter notebooks demonstrating applications for fintech clients.

Software Engineering Intern

Remote / UK

Microsoft

Jul – Aug 2021

- Built and documented an automated monitoring solution for critical infrastructure.
- Acted as technical lead among interns—wrote high-level design docs, test plans, and conducted code reviews.

Publications

Journal Articles

Personalized MR-Informed Diffusion Models for 3D PET Image Reconstruction

Aug 2025

G. Webber et al., *IEEE Transactions on Radiation and Plasma Medical Sciences*

[doi:10.1109/TRPMS.2025.3602262](https://doi.org/10.1109/TRPMS.2025.3602262) 

Likelihood-Scheduled Score-Based Generative Modeling for Fully 3-D PET Image Reconstruction [G. Webber](#) et al., *IEEE Transactions on Medical Imaging*

June 2025

[doi:10.1109/TMI.2025.3576483](https://doi.org/10.1109/TMI.2025.3576483) 

Diffusion Models for Medical Image Reconstruction

Sept 2024

[G. Webber](#) & A. J. Reader, *BJR—Artificial Intelligence*

[doi:10.1093/bjrai/ubae013](https://doi.org/10.1093/bjrai/ubae013) 

Conference Proceedings

Steerable Conditional Diffusion for Domain Adaptation in PET Image Reconstruction

To appear: Nov 2025

[G. Webber](#) et al., *IEEE Medical Imaging Conference (MIC) 2025* (accepted oral), Yokohama, Japan

Supervised Diffusion-Model-Based PET Image Reconstruction

To appear: Sept 2025

[G. Webber](#) et al., *MICCAI 2025* (accepted poster), Daejeon, South Korea

[arXiv:2506.24034](https://arxiv.org/abs/2506.24034) 

Generative-Model-Based Fully 3-D PET Image Reconstruction by Conditional Diffusion Sampling

Nov 2024

[G. Webber](#) et al., *IEEE MIC 2024*, Tampa, USA (oral)

[doi:10.1109/NSS/MIC/RTSD57108.2024.10657861](https://doi.org/10.1109/NSS/MIC/RTSD57108.2024.10657861) 

Multi-Subject Image Synthesis as a Generative Prior for Single-Subject PET Reconstruction

Nov 2024

[G. Webber](#) et al., *IEEE MIC 2024*, Tampa (poster)

[doi:10.1109/NSS/MIC/RTSD57108.2024.10657446](https://doi.org/10.1109/NSS/MIC/RTSD57108.2024.10657446) 

Submitted Manuscripts

Distributional Consistency Loss: Solving Noisy Inverse Problems Without Overfitting

Submitted: May 2025

[G. Webber](#) & A. J. Reader, *NeurIPS 2025* (under review)

Technologies

Programming: Proficient with Python (NumPy, Pandas, SciPy; PyTorch, TensorFlow, scikit-learn; Matplotlib, Napari). Familiar with Java, C#, Scala, HTML, SQL, JavaScript, CSS, Haskell.

Tools & Workflow: Git / GitHub, Linux & Windows, Azure + GCS storage, Scrum / Agile (Confluence, Jira), LaTeX.

Activities

- Won sponsor prize at **Oxford Hack 2022** for novel LiDAR-data visualisation for Oxa autonomous vehicles.
- Elected **Student Representative** for the Smart Medical Imaging CDT cohort – organise research-development seminars and social events.